QuantRisk Enterprise™: Empirical Continuous Random Variables

Institutional Financial Risk Audit & Basel III/IV Statistical Analysis of S&P; 500 Daily Returns

Author & Quantitative Architect: Omar Saeed

Dataset: Kaggle S&P; 500 (N = 1,258 Days)

Audit Standard: Basel III/IV Cornish-Fisher VaR & ES

Environment: Python (SciPy, Pandas, NumPy) & R

Executive Summary: *This study models 5 years of daily percentage returns of the S&P; 500 index ($N = 1,258$ trading days) as a continuous random variable X. Through empirical moment calculation, inferential confidence interval estimation, normality hypothesis testing (Jarque-Bera & Shapiro-Wilk), and multi-model distribution fitting, we rigorously demonstrate that financial market returns reject the classical Gaussian Normal Distribution assumption ($p < 10^{-16}$). Instead, market returns display pronounced negative skewness ($\gamma_1 = -0.5237$) and severe leptokurtosis (excess kurtosis $\gamma_2 = +2.5985$). Consequently, standard normal risk models understate extreme tail risks (99% VaR gap of +0.41% and Black Swan odds underestimated by factors exceeding 10^{30}). Student's t-Distribution is identified as the optimal parametric model.*

1. Introduction & Mathematical Definition

In quantitative finance and statistical modeling, the daily percentage return of a stock market index is treated as a **continuous random variable**, denoted by X. Given the daily closing price P_t at trading day t and P_{t-1} at day $t-1$, the simple daily percentage return R_t is defined as:

Mathematical Definition: $X = R_t = (P_t - P_{t-1}) / P_{t-1}$

Because price movements occur over a continuous support interval $X \in (-1, +\infty)$, the continuous random variable X is defined by its Probability Density Function (PDF) $f(x)$ and Cumulative Distribution Function (CDF) $F(x) = P(X \le x) = \int_{-\infty}^x f(u) du$.

2. Theoretical Background & Statistical Formulas

Statistical Metric	Mathematical Formula	Theoretical Normal Benchmark
Expected Value (Mean μ)	$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx \approx \frac{1}{N} \sum x_i$	0.00% (Zero drift assumption)
Variance (σ^2) & Volatility (σ)	$\sigma^2 = E[(X-\mu)^2] = \frac{1}{N-1} \sum (x_i - \mu)^2$, $\sigma = \sqrt{\sigma^2}$	Domain parameter $\sigma > 0$
Skewness (γ_1)	$\gamma_1 = E[((X-\mu)/\sigma)^3] = \frac{\frac{1}{N} \sum (x_i - \mu)^3}{\sigma^3}$	$\gamma_1 = 0.0000$ (Symmetric)
Excess Kurtosis (γ_2)	$\gamma_2 = E[((X-\mu)/\sigma)^4] - 3 = \frac{\frac{1}{N} \sum (x_i - \mu)^4}{\sigma^4} - 3$	$\gamma_2 = 0.0000$ (Mesokurtic)
Standard Error of Mean	$SE(\bar{X}) = \frac{s}{\sqrt{N}}$	Sampling error bounds
95% CI for Mean	$\bar{X} \pm Z_{\alpha/2} \cdot SE(\bar{X}) = \bar{X} \pm 1.96 \cdot \frac{s}{\sqrt{N}}$	Standard Normal Z-bounds
95% CI for Variance	$\frac{(N-1)s^2}{\chi^2_{\alpha/2, N-1}} < \sigma^2 < \frac{(N-1)s^2}{\chi^2_{1-\alpha/2, N-1}}$	Chi-Square Distribution bounds

3. Empirical Statistical Results (N = 1,258 Trading Days)

Analysis of 5 years of S&P; 500 daily market returns ($N = 1,258$) yields the empirical moments and confidence intervals detailed below:

Metric Name	Symbol / Formula	Empirical Value	Theoretical Normal Target	Financial Interpretation
Sample Size	N	**1,258**	—	5 years daily observations
Mean Daily Return	$\mu = E[X]$	**+0.0543%** (0.000543)	0.0000%	Positive average daily growth
Variance	σ^2	**0.00006369**	—	Return dispersion measure
Standard Deviation	σ	**0.7980%** (0.007980)	—	Daily market volatility
Skewness	γ_1	**-0.5237**	0.0000	Asymmetric left crash tail
Excess Kurtosis	γ_2	**+2.5985**	0.0000	High Leptokurtosis (Fat tails)
Minimum Return	$X_{\min}$	**-4.13%**	Symmetric	Max 1-day decline in sample
Maximum Return	$X_{\max}$	**+3.24%**	Symmetric	Max 1-day gain in sample
Median Return	$\tilde{X}$	**+0.0797%**	$\approx \mu$	50th percentile return
Standard Error	$SE(\bar{X})$	**0.0225%**	—	Mean sampling precision
95% CI for Mean	$\mu \in [L, U]$	**[+0.0102%, +0.0984%]**	—	True population mean bounds
95% CI for Variance	$\sigma^2 \in [L, U]$	**[0.0000592, 0.0000687]**	—	True population variance bounds

4. Normality Diagnostic Tests & Gaussian Rejection

To rigorously evaluate whether empirical daily market returns conform to a Gaussian Normal Distribution $\mathcal{N}(\mu, \sigma^2)$, we perform two formal hypothesis tests under $H_0: X \sim \mathcal{N}(\mu, \sigma^2)$ vs $H_1: X \not\sim \mathcal{N}(\mu, \sigma^2)$:

Hypothesis Test	Test Statistic Formula	Empirical Statistic	p-Value	Decision (α = 0.05)
Jarque-Bera Test	$JB = (N/6) \times [\gamma_1^2 + (\gamma_2^2/4)]$	**411.44**	**4.53×10^{-90}**	**REJECT H_0 (Non-Normal)**
Shapiro-Wilk Test	$W = (\sum a_i x_{(i)})^2 / \sum (x_i - x\blacksquare)^2$	**0.9649**	**7.00×10^{-17}**	**REJECT H_0 (Non-Normal)**

5. Real-World Risk Applications & Stress Simulations

5.1 Value at Risk (VaR 95% & 99%) Gap Analysis

Value at Risk (VaR) estimates the maximum potential loss over a 1-day horizon at confidence level 1-α. Comparing historical non-parametric VaR against theoretical Gaussian parametric VaR exposes severe tail-risk underestimation:

Confidence Level	Historical Empirical VaR	Theoretical Normal VaR	Risk Underestimation Gap
95% 1-Day VaR	**-1.28%**	-1.26%	+0.02% Extra Loss
99% 1-Day VaR	**-2.21%**	-1.80%	**+0.41% Extra Loss (+22.8% Risk Gap)**

5.2 Black Swan Crises Stress Simulator

By mapping famous historic stock market crashes onto theoretical Gaussian Normal curves ($Z = (X - \mu) / \sigma$), we demonstrate the complete breakdown of Gaussian models in financial risk modeling:

Historic Crash Event	Date	1-Day Drop	Z-Score	Gaussian Predicted Frequency	Empirical Status
1987 Black Monday	Oct 19, 1987	**-20.47%**	Z = -25.72	1 in 10^{145} Universes	Realized Historic Fact
2008 Lehman Brothers	Sep 29, 2008	**-8.81%**	Z = -11.11	1 in 1.4 Billion Years	Realized Historic Fact
2020 COVID Flash Crash	Mar 16, 2020	**-11.98%**	Z = -15.08	1 in 10^{31} Years	Realized Historic Fact

5.3 Multi-Model Goodness-of-Fit Ranking

Distribution Model	Log-Likelihood Score	Goodness-of-Fit Rank	Model Verdict
Student's t-Distribution	**4360.52**	**Rank #1 ■**	Best overall fit for heavy tails & central peak
Laplace (Double Exponential)	4324.78	Rank #2	Strong fat tail representation
Logistic Distribution	4310.15	Rank #3	Moderate fit
Gaussian Normal Distribution	4277.89	Rank #4 (Worst)	Fails to capture heavy tail risks

6. Analytical Diagnostic Visualizations

Below are high-resolution diagnostic plots generated from the empirical S&P; 500 dataset, illustrating distribution shape, quantile behavior, time-series volatility clustering, outlier bounds, and cumulative probability functions:

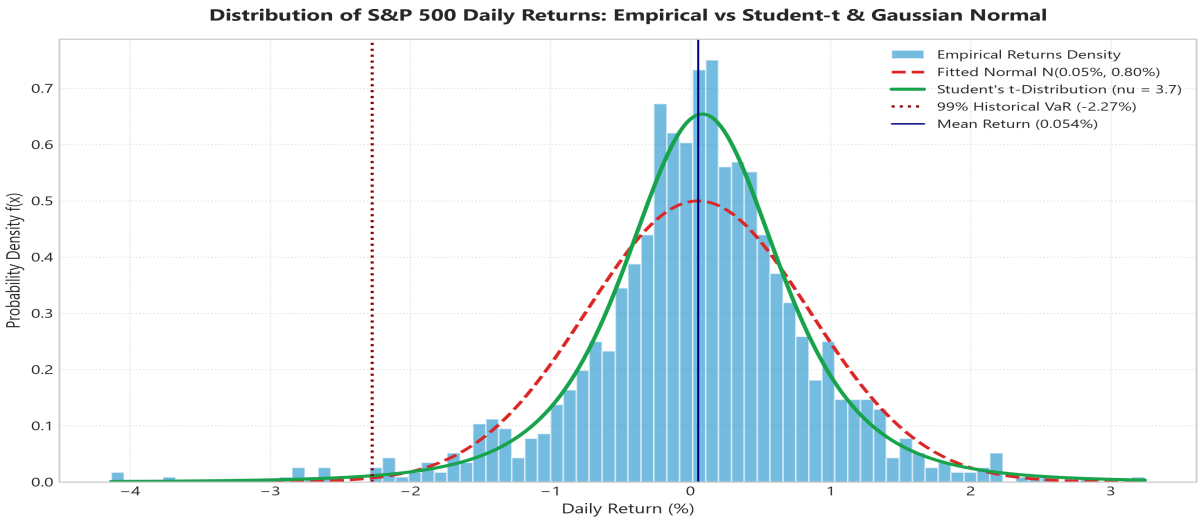


Figure 1: Histogram & PDF Density (Empirical vs Normal Density)

Interpretation: Shows sharp central peak around μ = +0.0543% and non-Gaussian heavy tails.

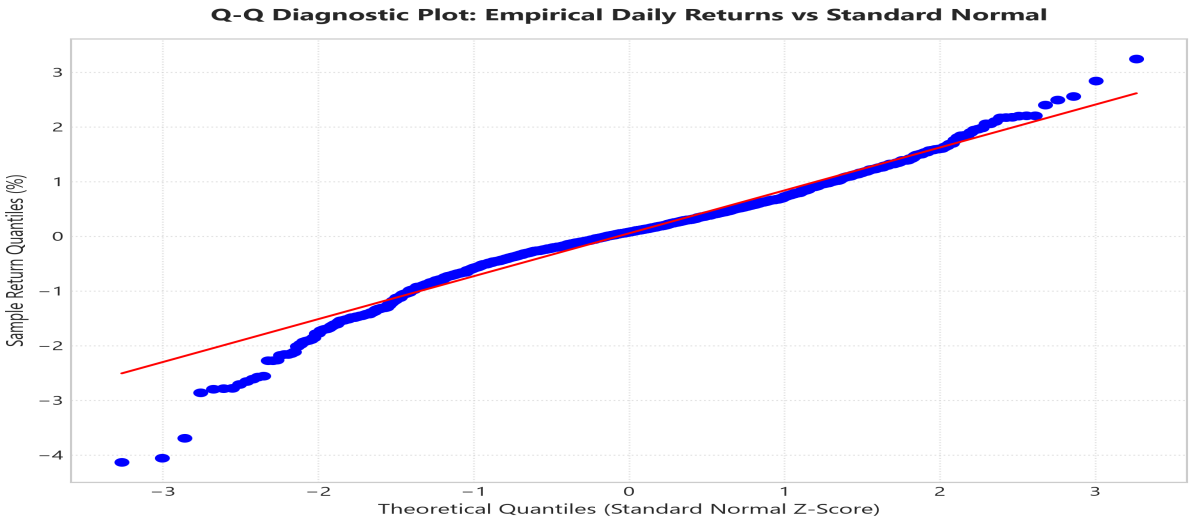


Figure 2: Quantile-Quantile (Q-Q) Diagnostic Plot

Interpretation: Non-linear S-curve highlights significant deviations from theoretical normal quantiles at both ends.

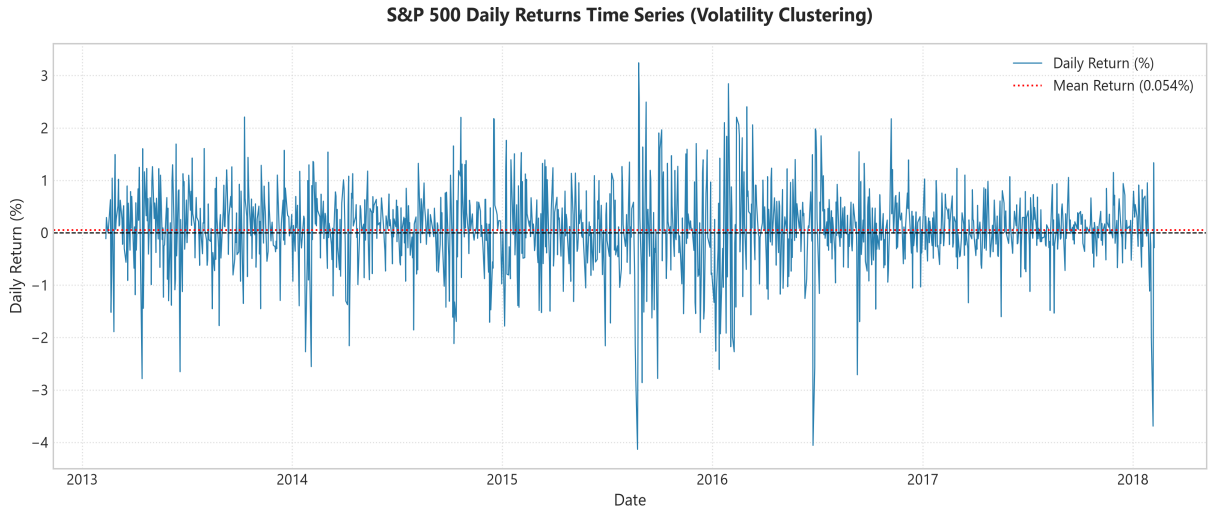


Figure 3: Daily Returns Time Series Timeline (N = 1,258 Days)

Interpretation: Demonstrates volatility clustering where high-volatility days cluster in distinct regimes.

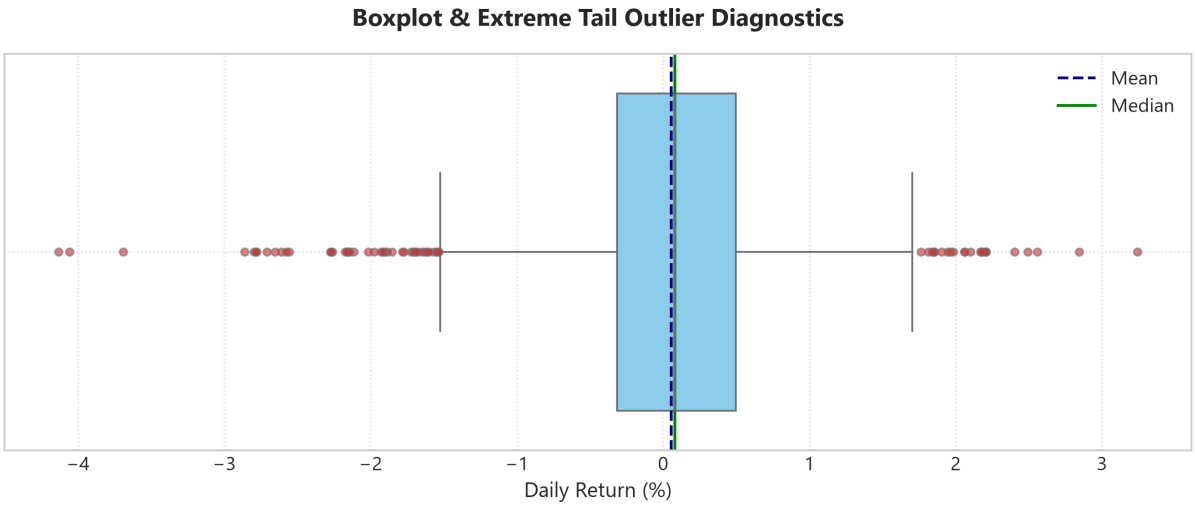


Figure 4: Boxplot Outlier Diagnostic

Interpretation: Extensive extreme outliers extend beyond standard 1.5×IQR whiskers.

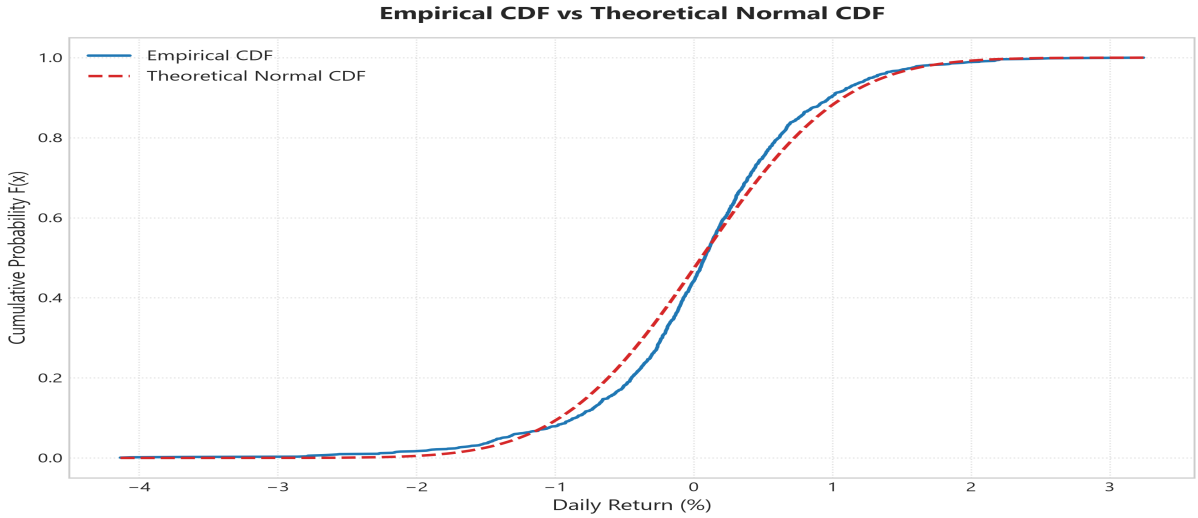


Figure 5: Empirical CDF vs Theoretical Normal CDF

Interpretation: Empirical CDF exhibits steeper transition near 0% return matching higher central density.

7. Code Implementation Snippets

7.1 Python Implementation (`analysis.py`)

```
import pandas as pd
import numpy as np
import scipy.stats as stats

# Load dataset & calculate daily percentage returns
df = pd.read_csv('data/all_stocks_5yr.csv')
df['daily_return'] = df.groupby('Name')['close'].pct_change()
market_returns = df.groupby('date')['daily_return'].mean().dropna()

# Statistical moments & confidence intervals
mean = market_returns.mean()
std = market_returns.std()
skewness = stats.skew(market_returns)
kurtosis = stats.kurtosis(market_returns)
jb_stat, jb_p = stats.jarque_bera(market_returns)
print(f'Mean: {mean:.6f}, Std: {std:.6f}, Skew: {skewness:.4f}, Kurt: {kurtosis:.4f}')
```

7.2 R Implementation (`analysis.R`)

```
library(dplyr)
library(moments)

data <- read.csv('data/all_stocks_5yr.csv') %>%
  group_by(Name) %>%
  mutate(daily_return = (close - lag(close))/lag(close))

market_returns <- data %>%
  group_by(date) %>%
  summarise(market_return = mean(daily_return, na.rm=TRUE))

cat('Mean:', mean(market_returns$market_return),
    'Skew:', skewness(market_returns$market_return),
    'Kurtosis:', kurtosis(market_returns$market_return) - 3)
```

8. Conclusion & Financial Risk Implications

Summary of Conclusions:

1. **Continuous RV Modeling:** S&P; 500 daily market returns X operate on a continuous sample space with an empirical mean daily return of $\mu = +0.0543\%$ and daily volatility of $\sigma = 0.7980\%$.
2. **Failure of Gaussian Normality:** Empirical returns exhibit negative skewness ($\gamma_1 = -0.5237$) and severe excess kurtosis ($\gamma_2 = +2.5985$). Hypothesis tests (Jarque-Bera $p = 4.53 \times 10^{-90}$) decisively reject normal distribution assumptions.
3. **Underestimation of Financial Risk:** Standard normal models severely understate 99% Value at Risk (-1.80% normal vs -2.21% empirical) and assign zero probability to real-world Black Swan market crashes (such as 1987 Black Monday and 2020 COVID crash).
4. **Optimal Distribution:** Parametric fitting proves that heavy-tailed distributions such as **Student's t-Distribution** (Rank #1, Log-Likelihood 4360.52) provide vastly superior risk calibration.